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1 Abstract

The neocortex is able to exhibit a remarkable ability to integrate sensory inputs, learn temporal regularities, and construct context-dependent representations that support prediction, abstraction, and generalisation. Biologically inspired cortical learning frameworks, particularly Hierarchical Temporal Memory (HTM) and the broader class of Cortical Learning Algorithms (CLAs), provide a principled computational account of these capabilities through sparse distributed representations (SDRs), active dendrites, and local plasticity mechanisms.

Despite their strengths in robust sequence learning and continual adaptation, existing CLA implementations remain largely limited to first-order temporal associations within individual cortical areas. They lack explicit mechanisms for representing higher-level contextual relationships such as semantic abstraction, multi-modal integration, and co-ordination across multiple cortical regions. This limitation constrains their ability to model contextual association as observed in biological cortex.

The aim of this research is to investigate how higher-level contextual association can be modelled within cortical learning frameworks by extending existing CLA architectures with mechanisms that support distributed, heterarchical interaction between multiple learning units. Rather than introducing supervised optimisation or global error signals, the research proposes architectural extensions that preserve sparse, unsupervised, and biologically plausible learning principles.

The proposed approach involves (i) analysing the representational and architectural constraints of current CLA and Neural Association Algorithm (NAA) models, (ii) investigating associative mechanisms in alternative biologically inspired models such as deep neural networks (DNNs) and spiking neural networks (SNNs), and (iii) developing an extended context-aware framework capable of encoding higher-order invariances, feature integration, cross-domain associations, and multi-modal temporal-spatial dependencies.

Core research questions include:

1. how to encode contextual dependencies beyond first-order temporal transitions
2. which architectural extensions enable hierarchical and semantic association layers, and
3. how the extended CLA compares to DNN/SNN-based associative mechanisms in terms of efficiency, generalization, and biological plausibility.

The expected outcome of this work is a technically grounded extension of cortical learning algorithms that demonstrates how distributed organisation and voting-based integration can support contextual association beyond simple temporal sequence memory. This contribution aims to clarify the conditions under which heterarchical structure improves contextual learning while maintaining the biological plausibility and continual learning properties of cortical-inspired models.

1.1 Structure of the document

This document is structured to progressively develop the theoretical motivation, methodological framework, and proposed contributions of the research.

Chapter 2 introduces the background and motivation for the work, providing an overview of cortical learning algorithms, the Neural Association Algorithm, and neuroscientific perspectives on contextual association. Chapter 3 presents a critical review of the relevant literature, examining cortical learning models, neuroscientific evidence, and related computational approaches, and identifies the research gap addressed by this project.

Chapter 4 defines the research aim, questions, and objectives that guide the investigation. Chapter 5 describes the overall methodology, including baseline models, proposed architectural extensions, implementation strategy, and evaluation approach. Chapter 6 discusses the progress and exploratory results obtained to date, highlighting their implications for subsequent phases of the research.

Chapter 7 outlines the anticipated originality and contribution to knowledge, while Chapter 8 presents the research plan and timeline. Chapter 9 describes the data management strategy, and Chapter 10 concludes the report by summarising the current status and direction of the PhD research.

2 Introduction and Background

This chapter introduces the key concepts and model families that form the foundation for the research presented in this document. Its purpose is to provide a high-level orientation to cortical learning algorithms, associative learning models, and computational approaches to context. A critical analysis of these approaches, including their limitations and open challenges, is provided in Section 3.

2.1 Cortical Learning Algorithms

Cortical Learning Algorithms (CLAs) are a class of biologically inspired computational models designed to capture core principles of neocortical processing, including sparse activity, local synaptic plasticity, and continual learning from streaming input. These models are motivated by anatomical and physiological observations suggesting that the neocortex is composed of repeated canonical circuits that learn structure from sensory experience without explicit supervision [17, 11].

Hierarchical Temporal Memory (HTM) is a prominent instance of this approach and has been proposed as a computational framework for modelling neocortical function [17]. The Cortical Learning Algorithm (HTM-CLA) formalises this framework using sparse distributed representations (SDRs), active dendrites, and Hebbian-like learning rules [16, 1]. Within HTM-CLA, information is encoded using high-dimensional sparse binary

vectors, enabling robustness to noise and overlap-based similarity comparisons.

The HTM-CLA architecture includes two principal components: the Spatial Pooler, which learns stable sparse encodings of feedforward input, and the Temporal Memory, which models transitions between successive representations and supports sequence learning and prediction [6]. These mechanisms provide a computational basis for learning temporal structure under continual, unsupervised conditions.

2.2 Neural Association Algorithm

The Neural Association Algorithm (NAA) has been proposed as a generalisation of cortical learning principles toward associative learning across distributed neural populations [8]. Building on the foundations of HTM-CLA, the NAA emphasises the formation of associations between sparse representations through dendritic processing and synaptic plasticity mechanisms inspired by cortical microcircuits.

In contrast to sequence-focused models, NAA introduces mechanisms that extend associative learning beyond pure temporal sequence memory by allowing contextual modulation within canonical units. Canonical units are modelled with multiple dendritic compartments, enabling different forms of contextual input to influence neuronal activation. Structural plasticity mechanisms regulate synaptic growth and removal, supporting stable yet adaptable formation of associations.

The NAA provides one example of prior work exploring how cortical-inspired architectures may extend beyond simple temporal sequence learning. Its detailed analysis, strengths, and limitations are examined in Chapter 3 alongside other cortical learning approaches.

2.3 Context in Neural and Computational Models

In neuroscience and cognitive science, context broadly refers to latent information that modulates perception, memory, and behaviour based on spatial, temporal, semantic, and task-related factors. Contextual representations are not localised to a single brain region but emerge from interactions among hippocampus, association cortex, and sensory areas [32, 30].

From a computational perspective, context is often operationalised as information that disambiguates otherwise similar inputs or guides prediction and inference over time. Different modelling paradigms capture context using distinct mechanisms, such as internal state variables, feedback signals, or distributed population activity. These perspectives motivate the investigation of architectures capable of integrating multiple contextual dimensions in a continual and biologically plausible manner.

2.4 Machine Learning Approaches to Context Modelling

Several established machine learning paradigms incorporate mechanisms for representing contextual information. Recurrent Neural Networks (RNNs) model temporal context through recurrent hidden states, enabling sequence learning over time [19]. Variants such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) introduce gating mechanisms to support longer temporal dependencies [6].

Attention-based architectures, particularly Transformers, represent context through attention mechanisms that weight relationships among input elements across a sequence [36]. These models have achieved strong empirical performance in a range of sequence and language tasks.

Spiking Neural Networks (SNNs) represent information using spike timing and firing dynamics, incorporating biologically inspired learning rules such as spike-timing-dependent plasticity [23]. SNNs offer a closer correspondence to biological neuronal dynamics and are often studied in the context of neuromorphic computing.

These approaches differ substantially in their learning rules, representational formats, and assumptions about supervision and optimisation. Their relationship to cortical-inspired learning models and their suitability for contextual association are examined in detail in Chapter 3.

2.5 Summary

This chapter has attempted to introduce the principal concepts and computational frameworks relevant to the research, including cortical learning algorithms, associative learning models, and alternative machine learning approaches to context modelling. The discussion has been intentionally descriptive, focusing on established ideas and prior work. Chapter 3 builds on this foundation by critically analysing these approaches, identifying their limitations, and motivating the research direction pursued in this document.

3 Literature Review

3.1 Introduction to the Literature Review

The aim of this literature review is to examine existing scientific and computational approaches relevant to the development of higher-level contextual association in cortical learning systems. While the preceding chapter introduced the biological motivation and outlined foundational models such as Hierarchical Temporal Memory (HTM) and the Neural Association Algorithm (NAA), this section evaluates the current state of research across neuroscience, artificial intelligence, cortical learning algorithms, and modern machine learning. Such critical evaluation is necessary to identify methodological limitations, and theoretical inconsistencies that justify the research direction pursued in this project.

To establish a coherent foundation for the research, the review is structured around three complementary strands of prior work. First, the literature on cortical learning algorithms is examined, with a focus on Hierarchical Temporal Memory (HTM), the Cortical Learning Algorithm (CLA), and related extensions that model sequence learning, sparse distributed representations, and local plasticity. Parallel to this, extensive neuroscientific evidence demonstrates that context in the brain is multidimensional, distributed, and supported by coordinated interactions across hippocampus, medial temporal cortex, prefrontal cortex, and sensory association areas [32, 5, 29]. Mechanisms such as feature binding [35], predictive coding [28], and cortico-hippocampal reinstatement highlight the richness of contextual processing but do not directly translate into computational models capable of continual learning. The third part of this literature review includes computational approaches to context in machine learning. Recurrent neural networks [19], attention-based models such as Transformers [36], predictive coding architectures such as hierarchical predictive coding [10] offer multiple mechanisms for encoding temporal and semantic dependencies. However, these models typically require global optimisation, large labelled datasets, or externally imposed memory structures. They also struggle with continual adaptation and catastrophic forgetting [21], limiting their capacity to form stable, biologically plausible contextual associations.

By bringing together these three domains, the literature review seeks to clarify how context has been conceptualised and modelled across different scientific traditions. The review pays particular attention to four dimensions of context—temporal, spatial, semantic, and functional—and evaluates the extent to which existing models support hierarchical or heterarchical integration of contextual signals. Through this critical synthesis, the review aims to establish the central research gap addressed by the present project: the absence of a computational framework that enables multiple distributed cortical units to form, exchange, and utilise higher-level contextual associations using sparse, continually learned representations. This gap forms the basis for the research aims, methodology, and contributions presented in the following sections.

3.2 Cortical Learning Algorithms (HTM-CLA)

3.2.1 Overview of the CLA Framework

HTM-CLA is a biologically inspired computational framework proposed in the literature as a model of neocortical computation [17, 16]. CLA proposes that cortical computation is driven by sparse distributed representations (SDRs), active dendrites, and local synaptic plasticity, enabling online learning of temporal patterns without global optimisation. Its architecture is composed of two major components: the Spatial Pooler (SP), which encodes stable sparse patterns from noisy input, and the Temporal Memory (TM), which learns transitions between successive SDRs and hence supports sequence prediction.

In the HTM-CLA framework, each cortical column is specified as containing multiple cells (commonly 32), allowing different temporal contexts to be represented for the same feedforward input [17]. Proximal dendrites encode feedforward sensory features, while distal and apical dendrites are used to incorporate lateral and top-down contextual signals, consistent with the HTM neuron model [6]. Information is represented using Sparse Distributed Representations (SDRs), which provide robustness to noise and enable similarity comparisons through overlap-based measures [1]. These principles align with experimental findings on sparse activity in cortical pyramidal cells, pattern completion, and temporal prediction [41, 26].

3.2.2 Strengths of HTM-CLA

CLA offers several computational advantages compared to mainstream machine learning approaches. First, it supports continual, online learning using Hebbian-like local rules, avoiding the catastrophic forgetting found in gradient-based systems [21]. Second, the use of SDRs allows high-capacity, noise-robust representations that preserve semantic similarity [1]. Third, Temporal Memory provides a powerful mechanism for learning and predicting sequences at multiple timescales using active dendrites and context-specific cell activation [6]. These properties collectively give CLA a form of temporal contextual memory that is biologically grounded and computationally efficient.

Moreover, empirical evaluations have demonstrated CLA’s strengths in anomaly detection, online prediction, and sequence modelling in non-stationary environments [22]. Its reliance on local dendritic processing and sparse coding aligns with observed cortical microcircuits, making it one of the few models that explicitly incorporate both structural and functional aspects of pyramidal neurons.

3.2.3 Limitations of HTM-CLA

A fundamental limitation of the CLA is that its sequence learning mechanism is primarily based on local transitions between successive sparse distributed representations, resulting in behaviour that closely resembles first-order or limited-order temporal modelling [12, 6].

While TM can handle branching sequences and multiple contexts, context is still defined by temporal adjacency rather than by semantic, spatial, or multi-modal associations. CLA lacks mechanisms for:

- forming relationships between non-adjacent events,
- integrating context across different sensory modalities,
- representing abstract or hierarchical concepts, and
- coordinating information between multiple cortical regions.

A major limitation arises from the empirical behaviour of the Spatial Pooler. While the SP is effective at generating sparse, stable patterns, its static receptive fields and fixed boosting mechanisms limit the formation of dynamic feature-based context, especially in complex, multi-modal tasks. Comparisons against modern unsupervised encoders (e.g., variational autoencoders or contrastive encoders) show that SP representations lack flexibility, abstraction, and hierarchical depth [24].

Furthermore, CLA’s architecture models only a fragment of the cortical circuit, primarily the superficial pyramidal layer (layer 2/3) and parts of layer 4. The deeper layers (5 and 6), thalamocortical loops, and heterarchical feedback pathways are only minimally represented or entirely absent. These omissions limit CLA’s ability to model top-down context modulation, attentional gating, and cross-area coordination [2].

3.2.4 Lack of Heterarchical Integration and Multi-Area Context

Recent neuroscience suggests that cortical processing is not purely hierarchical but involves extensive *heterarchical* (i.e., non-strict, recurrent organisation without a single top-down ordering) connectivity, where multiple areas jointly contribute to the representation of context and meaning [14]. CLA, however, does not include mechanisms for distributed consensus or voting-based interpretation across multiple cortical units.

This limitation is particularly evident when considering theories such as the Thousand Brains Theory (TBT), which propose that perception and intelligence emerge from thousands of parallel cortical models that reach agreement via voting and shared reference frames.[15] Current CLA implementations do not implement such voting structures, nor do they allow independent cortical modules to learn different aspects of the same object or context.

3.2.5 Summary

Following from the studies reviewed, HTM-CLA provides a biologically grounded and computationally efficient model of temporal sequence learning, but remains limited in its ability to support higher-level contextual association. Its constraints include the absence

of multi-area communication, lack of heterarchical organisation, and inability to form semantic or multi-modal contextual links. These limitations underscore the need for extended cortical learning frameworks—such as NAA or the heterarchical multi-unit architecture proposed in this research—that can incorporate distributed association, structured voting, and emergent context integration.

3.3 Neural Association Algorithm (NAA)

The Neural Association Algorithm (NAA) proposed by Dobric [8] is one of the first attempts to extend the classical Hierarchical Temporal Memory Cortical Learning Algorithm (HTM-CLA) toward a more general framework capable of modelling associations beyond first-order temporal transitions. Whereas HTM-CLA emphasises sequence learning, NAA focuses on the formation, strengthening, and utilisation of associations across distributed sparse neural populations, drawing on biological principles of dendritic processing, structural plasticity, and network-level context integration. The approach builds directly on the SDR theory of neuronal representation introduced by Ahmad and Hawkins [16, 1] and incorporates elements of cortical microcircuit organisation identified in biological research [14, 2].

3.3.1 Architectural Foundations of NAA

At the core of NAA is the concept of a *canonical cortical unit*—a computational module inspired by the operation of pyramidal neurons with multiple dendritic compartments. Distal dendrites encode lateral context, whereas apical dendrites provide feedback signals from higher-order areas. This design is grounded in empirical findings showing that cortical neurons integrate context through distinct dendritic zones [14]. By modelling synaptic permanence, growth, and removal, NAA aims to simulate the formation of stable but adaptable associations between SDR patterns, addressing the stability–plasticity problem that commonly affects gradient-based systems [21].

Unlike HTM-CLA—which limits context to temporal adjacency—NAA supports the formation of context-sensitive associations that do not necessarily depend on sequence order. This flexibility aligns with neuroscientific evidence showing that context is encoded across distributed cortico–hippocampal networks [5, 29] and that feedback loops contribute to predictive processing [2, 28, 10]. By incorporating these ideas, NAA extends the representational capacity of cortical learning algorithms beyond the predictive sequence memory of the Temporal Memory model.

3.3.2 Association Mechanisms in NAA

The key underlying mechanism NAA is the formation of associative connections between active SDR patterns across time, space, or functional co-occurrence. These associations allow canonical units to develop internal relational structures that support

context-dependent behaviour. Unlike recurrent neural networks [19, 4] or spiking systems based on STDP [23, 33], NAA does not rely on global optimisation or precise spike timing. Instead, associations are learned incrementally through local synaptic updates, consistent with biological cortical operation and in stark contrast to models that require backpropagation-through-time and suffer from vanishing gradients [3].

Dobric introduces a *newborn stage* for canonical units, regulated through homeostatic plasticity, in which mini-columns are initially encouraged to participate uniformly in learning before competitive dynamics stabilise. [8] This mechanism is proposed to improve learning stability and to reduce representational bias during early stages of training.

From a biological perspective, this design choice is motivated by observations of developmental overproduction and subsequent pruning in cortical circuits, where early broad connectivity is progressively refined through experience [14]. In contrast to this locally regulated developmental process, most deep learning systems rely on replay or rehearsal strategies to mitigate catastrophic forgetting during sequential learning.

3.3.3 Limitations and Challenges of NAA

Despite its promise, NAA exhibits several limitations that constrain its ability to model high-level contextual association. First, while NAA can encode associations across distributed SDRs, it still operates primarily within a single cortical unit or a small set of weakly interacting units. This contrasts with neuroscientific evidence showing that context is distributed across large-scale networks involving hippocampus, prefrontal cortex, and sensory areas [32, 30, 43]. NAA lacks an explicit mechanism for large-scale coordination or multi-area integration.

Second, NAA does not implement a heterarchical or voting-based consensus mechanism. This is a crucial gap given theories such as the Thousand Brains Theory, which argues that perception arises from multiple parallel models voting on object identity and context.

Third, although NAA generalises associative learning beyond pure sequence memory, it does not yet provide an explicit, architecturally defined mechanism for systematically representing semantic, spatial, or functional context in a unified representation. Neuroscience studies have shown that context involves feature binding [35, 40], predictive modulation [37], and multi-modal integration, all processes that require interaction across multiple specialised cortical units rather than within a single module.

3.3.4 Critical Evaluation

NAA represents a substantial extension of the HTM-CLA framework by providing mechanisms for associative learning that are biologically grounded, sparse, and continual. Its use of dendritic compartments, SDR-based association, and homeostatic control addresses limitations in both deep learning [36, 39, 18] and classical cortical algorithms. However,

NAA remains constrained by the absence of multi-area coordination, heterarchical routing, and structured voting—features that appear essential for modelling higher-order contextual association in biological cortex. These limitations motivate the development of extended architectures, such as the heterarchical multi-unit framework proposed in this research, which aim to support distributed contextual integration across multiple cortical modules.

3.4 Comparative Analysis of DNNs, SNNs, and Cortical Learning Algorithms

Following the overview in Section 3.1, this section provides a comparative analysis of deep neural networks, spiking neural networks, and cortical learning algorithms as alternative computational approaches to contextual association. Deep Neural Networks (DNNs), Spiking Neural Networks (SNNs), and Cortical Learning Algorithms (CLAs) represent three distinct paradigms for modelling learning and association in neural systems. Although all three are, in principle, computationally universal, they differ substantially in their learning rules, representational formats, temporal dynamics, biological plausibility, and suitability for continual and contextual learning [23, 19, 16, 6, 1].

3.4.1 Learning Paradigms and Training Signals

DNNs rely primarily on gradient-based optimisation (backpropagation-through-time in recurrent settings), using a global error signal to update synaptic weights across many layers. This yields high task performance but requires large labelled datasets and repeated passes over static training sets [19, 4, 36]. SNNs, in contrast, typically employ local learning rules such as spike-timing-dependent plasticity (STDP) and its variants, which adjust synapses based on relative spike timing and occasionally additional modulatory signals [23, 33, 42]. CLAs, including HTM-CLA, use local Hebbian-like rules operating on sparse binary activations and synaptic permanence values, supporting purely online, unsupervised adaptation without explicit error backpropagation [16, 6].

From the perspective of contextual association, DNNs learn associations implicitly as a by-product of error minimisation; recurrent and attention-based architectures encode temporal and relational context in their hidden states and attention weights but remain dependent on global supervision [19, 36]. SNNs and CLAs instead rely on self-organising local updates that directly link co-active patterns, which is more compatible with continual learning but offers weaker guarantees of task-optimality [21, 1].

3.4.2 Representational Formats and Context Encoding

DNNs generally employ dense, real-valued distributed representations, in which many units participate in coding any given feature. Recurrent networks and Transformers encode context in continuous state vectors or attention maps, enabling rich temporal

and semantic dependencies but at the cost of opaque, high-dimensional representations [19, 36]. SNNs represent information in the timing and rate of spikes; recurrent spiking circuits, reservoir SNNs, and neuromorphic architectures exploit temporal dynamics to encode context over short to medium time scales [23, 42].

CLAs explicitly use Sparse Distributed Representations (SDRs), in which a small fraction of neurons are active in a high-dimensional binary vector [1]. SDRs provide combinatorial representational capacity, noise robustness, and a simple overlap-based similarity metric that can be exploited for associative retrieval and anomaly detection [16]. In comparison to DNNs and SNNs, CLA’s deliberately sparse, binary coding makes contextual similarity and novelty explicitly measurable, but current implementations primarily encode temporal rather than semantic or multi-modal context.

3.4.3 Biological Plausibility and Alignment with Cortical Microcircuits

DNNs are biologically inspired at a very abstract level but diverge markedly from known cortical mechanisms, especially due to the use of exact weight symmetry, global error signals, and non-local credit assignment [3, 21]. SNNs, by incorporating discrete spikes and synaptic plasticity based on spike timing, are much closer to biophysical neuronal models and are often implemented on neuromorphic hardware designed to mimic event-driven cortical computation [23, 42].

CLAs occupy an intermediate level of abstraction. They model cortical columns, active dendrites, local synaptic plasticity, and sparse activation in a way that is consistent with canonical microcircuit theories [14, 2, 16]. Distal dendritic segments capture contextual input from other neurons, while proximal segments encode feedforward sensory patterns, aligning with anatomical findings on pyramidal cell integration. However, CLAs abstract away precise spike timing and many biophysical details, focusing instead on an algorithmic account of sequence learning and prediction.

3.4.4 Computational Efficiency, Hardware, and Continual Learning

DNNs achieve state-of-the-art accuracy on many benchmarks but are computationally intensive, typically running on GPUs/TPUs and consuming significant energy, particularly for large Transformer models [36, 39]. Catastrophic forgetting remains a central problem in continual learning settings. SNNs can, in principle, match DNN performance on some tasks while using far fewer operations, especially when deployed on neuromorphic hardware, thanks to sparse event-driven computation [23, 42]. Nevertheless, training deep SNNs remains challenging and often relies on surrogate gradients or ANN-to-SNN conversion.

CLAs are computationally efficient in a different sense: operations are sparse, binary, and local, and the same network can be used for training and inference in an online manner. HTM-based systems have shown strong performance for streaming anomaly

detection and non-stationary time series prediction, where continual adaptation and robustness to distributional shift are crucial [22]. However, CLAs lack the mature hardware ecosystem of DNNs and SNNs, and have not been scaled to the same degree on large supervised benchmarks.

3.4.5 Suitability for Higher-Level Contextual Association

An important dimension for comparing learning paradigms concerns their capacity to support higher-level contextual association, particularly the integration of temporal, spatial, semantic, and functional information. DNNs, especially Transformers, provide powerful mechanisms for long-range temporal and relational dependencies via self-attention, but they depend on offline training, dense global states, and are not naturally suited for life-long, incremental learning [36, 39]. SNNs offer rich temporal dynamics and biologically grounded association via STDP, but current models typically lack structured mechanisms for semantic abstraction and large-scale context integration beyond local circuits [23, 33].

CLAs, and NAA as an extension, offer a principled framework for sparse, robust, and interpretable temporal context encoding [16, 1, 8]. However, they remain limited to single- or few-region architectures, and do not yet implement heterarchical multi-area coordination or explicit voting-based consensus across multiple cortical units. This comparative analysis therefore reinforces the central research gap identified earlier: no existing paradigm provides a cortical-inspired, heterarchical architecture in which multiple CLA/NAA-like modules form, share, and exploit higher-level contextual associations in a continual, biologically plausible manner. Addressing this gap motivates the heterarchical, voting-based cortical framework proposed by the present research.

3.4.6 Comparative Analysis

Table 1 summarises the comparative analysis of deep neural networks, spiking neural networks, and cortical learning algorithms presented in Section 3.4.

Dimension	Deep Neural Networks (DNNs)	Spiking Neural Networks (SNNs)	Cortical Learning Algorithms (CLA)
Learning Mechanism	Global optimisation via backpropagation; requires large datasets; prone to catastrophic forgetting.	Local learning via STDP or spike-based rules; supports event-driven adaptation.	Local Hebbian-like updates; online, unsupervised continual learning; no backpropagation.
Representation Format	Dense continuous vectors; high-dimensional and entangled; hard to interpret.	Spike timing and firing rate encode information.	Sparse Distributed Representations (SDRs); overlap-based similarity.
Context Encoding	Temporal/semantic context via hidden states or attention.	Temporal context via spike dynamics; limited abstraction.	Temporal predictive context via active dendrites; weak semantic/multi-modal context.
Biological Plausibility	Low (global error signals, weight symmetry).	Medium-high (event-driven, local updates).	Intermediate (columnar structure, dendrites, SDRs).
Computational Efficiency & Continual Learning	Compute-heavy; poor continual learning without replay.	Energy-efficient on neuromorphic hardware; training difficult.	Sparse, binary, efficient; excellent continual learning.

Table 1: Comparison of DNNs, SNNs, and CLAs across key computational dimensions.

The comparison in Table 1 highlights systematic differences between deep neural networks, spiking neural networks, and cortical learning algorithms with respect to learning mechanisms, representational formats, biological plausibility, and suitability for continual learning. While deep neural networks and attention-based architectures demonstrate strong performance on large-scale supervised tasks, they rely on dense representations, global optimisation, and offline training procedures. Spiking neural networks offer closer alignment with biological dynamics and energy-efficient computation, but current models remain limited in their ability to support higher-level abstraction and large-scale contextual integration.

Across these families of models, two broad limitations remain unresolved. First, context is usually treated as an external variable—such as a hidden state, attention vector, or latent code—rather than emerging from distributed population dynamics, as observed in cortical circuits. Second, most architectures are not designed for continual, online learning and therefore suffer from catastrophic forgetting [21], making them unsuitable for forming long-term, hierarchical, and multi-modal contextual associations without re-training or replay.

Cortical learning algorithms occupy a distinct position in this comparison. Their use of sparse distributed representations, local plasticity, and online learning provides strong robustness and continual adaptation, but existing implementations primarily encode temporal context and lack mechanisms for integrating semantic, spatial, or multi-modal information across multiple learning units. This comparison clarifies that no existing paradigm simultaneously satisfies the requirements of biological plausibility, continual learning, and higher-level contextual association. These observations motivate further investigation into architectural extensions of cortical learning models, which are examined in the subsequent sections.

3.5 Unsupervised Learning in Cortical Algorithms: Principles, Advantages, and Limitations

A defining characteristic of cortical learning algorithms such as HTM-CLA and the Neural Association Algorithm is their reliance on purely unsupervised, local learning rules. This stands in sharp contrast to most deep learning architectures, which depend on global error signals, large labelled datasets, and iterative optimisation procedures [19, 4, 36]. The biological cortex receives no explicit labels for the vast majority of its sensory input; learning proceeds through exposure to streams of unstructured, continuously changing data [17, 11, 26]. In this respect, cortical models must operate under conditions much closer to unsupervised continual learning than supervised gradient descent.

HTM-CLA embodies this principle through mechanisms such as sparse distributed representations (SDRs), Hebbian-like synaptic updates, and active dendritic processing [16, 1]. Learning occurs whenever neurons co-activate within specific temporal contexts, enabling the Temporal Memory to acquire predictive models of sequences without explicit teaching signals. This unsupervised paradigm allows CLA models to adapt online, respond to distributional shifts, and maintain robustness in streaming non-stationary environments [22]. These features address several limitations of supervised architectures, particularly their susceptibility to catastrophic forgetting, their dependence on offline training cycles, and the opacity of their dense distributed representations.

From a neuroscientific perspective, unsupervised learning is consistent with evidence that cortical circuits extract statistical regularities, build predictive models of the environment, and integrate multi-source context without labelled supervision [14, 2, 32]. Sparse population codes, generative inference, and feedback-modulated prediction errors contribute to the brain’s ability to learn context-rich structures from naturalistic input [28, 10]. These processes rely heavily on local dendritic integration and plasticity, features that HTM-CLA and NAA attempt to capture algorithmically.

However, the unsupervised nature of CLA also introduces significant limitations that directly inform the research gap of this project. While CLA effectively learns temporal transitions, it does not infer higher-order associations such as semantic relationships, spatial regularities, or multi-modal contextual bindings. Existing implementations operate primarily within a single cortical region, lacking mechanisms for coordination, communication, or consensus across multiple independent learning modules. Neuroscience research indicates that contextual representation arises from distributed networks involving hippocampus, prefrontal cortex, and association areas [5], suggesting that unsupervised learning at the local columnar level must be complemented by network-level interaction.

NAA partially addresses this gap by introducing dendritic association mechanisms and local structural plasticity, enabling context-sensitive links that go beyond simple temporal adjacency. Yet even NAA remains limited by the absence of heterarchical routing or a voting mechanism across multiple cortical units, features that appear critical for higher-level contextual inference [15]. Thus, while unsupervised learning is founda-

tional to biological and cortical-inspired models, its current algorithmic implementations are insufficient for modelling the full richness of contextual associations observed in the brain.

The importance of unsupervised learning therefore lies in both its advantages and its limitations: it provides the correct biological foundation for continual learning, but requires architectural extensions—such as distributed SDR voting, multi-area integration, and heterarchical organisational principles—to support higher-level contextual association. Addressing these limitations forms a core component of the present research.

3.6 Heterarchical Processing and Voting in Cortical Networks

Classical models of cortical organisation emphasised a largely feedforward hierarchy from sensory to associative areas [9]. However, anatomical and physiological evidence now suggests that cortical processing is better described as a *heterarchical* network with extensive recurrent, lateral, and feedback connections [14, 2]. Feedforward and feedback pathways form loops across multiple levels, while horizontal connections link neighbouring columns and areas, allowing information to circulate and be refined rather than simply propagated upward. This architecture implies that perception, memory, and context emerge from the joint activity of distributed regions rather than from a single, strictly layered pathway.

Within this heterarchical network, many studies have argued that cortical populations encode probability distributions and combine evidence in a manner analogous to voting or consensus formation. Population coding frameworks show how different neuronal groups can represent alternative hypotheses, with downstream activity reflecting a weighted combination of these signals [27]. In decision-making circuits, activity in parietal and prefrontal areas accumulates evidence from multiple sensory and associative sources until a threshold is reached, consistent with distributed consensus dynamics rather than a single locus of control [13, 20]. Similarly, cortico-hippocampal models of episodic memory suggest that different regions contribute complementary aspects of context, and successful retrieval reflects convergence across these networks [5, 29].

Predictive coding and related theories provide a functional interpretation of this architecture, proposing that cortical microcircuits implement recurrent message passing between levels, where top-down predictions and bottom-up prediction errors are combined to infer the most likely causes of sensory input [28, 10, 2]. In this view, each area maintains its own generative model, and coherent perception arises from iterative negotiation among these models. The process can be understood as a form of graded voting, in which different regions propose predictions and adjust them based on mismatches with incoming evidence and with each other [38, 31]. Feature binding and attention studies likewise indicate that object representations result from the coordinated selection and integration of distributed feature maps, guided by both bottom-up salience and top-down goals [35, 40].

The Thousand Brains Theory (TBT) makes this intuition explicit by proposing that

thousands of cortical columns learn separate models of objects and their locations, with perception emerging from a voting process across these columns [15]. Each column develops its own partial, viewpoint-specific model, and agreement across many columns yields a stable object or context representation. Although current implementations such as multi-region HTM architectures remain preliminary, the underlying idea is consistent with empirical evidence for distributed, partially redundant coding across cortical areas [14, 37].

From the perspective of contextual association, these findings suggest that higher-level context is not encoded in a single “context module” but emerges from interactions among multiple specialised regions, each contributing its own constraints and predictions. Heterarchical connectivity enables information to flow along multiple paths, while population-based coding and predictive message passing provide mechanisms for combining these signals in a voting-like manner. For cortical learning algorithms, this motivates moving beyond single-area models toward architectures in which multiple CLA/NAA-like units form their own unsupervised representations and arrive at shared contextual interpretations through explicit or implicit voting. The present project takes this perspective as a guiding principle for designing and evaluating extended cortical learning frameworks for higher-level contextual association.

4 Aim, Research Questions, and Objectives

The literature review in Chapter 3 highlights several complementary limitations across existing approaches to contextual learning, which together define the research gap addressed in this thesis.

First, cortical learning algorithms such as HTM-CLA provide biologically grounded mechanisms for sparse, unsupervised, and continual learning, but primarily encode context through temporal adjacency within individual cortical regions. As discussed in Section 3.2, these models lack explicit mechanisms for integrating semantic, spatial, or functional context, and provide only limited support for coordination across multiple cortical areas.

Second, extensions such as the Neural Association Algorithm broaden associative learning beyond pure sequence memory, yet remain constrained at the level of single or weakly coupled canonical units. Section 3.3 shows that, while NAA introduces dendritic association and structural plasticity, it does not offer an explicit architectural mechanism for unified, multi-dimensional contextual integration across distributed cortical modules.

Third, alternative computational approaches reviewed in Section 3.4 demonstrate complementary strengths and weaknesses. Deep neural networks and attention-based architectures are capable of capturing higher-order dependencies and abstraction, but rely on dense representations, global optimisation, and offline training, making them poorly suited to biologically plausible continual learning. Spiking neural networks offer

closer alignment with neural dynamics, but current models lack scalable mechanisms for higher-level contextual abstraction and coordinated multi-area inference.

Taken together, these findings reveal a clear research gap: the absence of a computational framework that combines sparse and unsupervised learning, continual adaptation, biological plausibility, and explicit architectural support for higher-level contextual association emerging from interaction across multiple distributed learning units. Addressing this gap motivates the research aims, questions, and objectives presented in the following sections.

4.1 Research Aim

The primary aim of this research is to investigate how higher-level contextual association can be modelled within cortical learning frameworks by extending existing cortical learning algorithms with heterarchical organisational principles and distributed voting mechanisms. The research seeks to move beyond first-order temporal sequence learning toward architectures capable of integrating temporal, semantic, spatial, and functional context in a biologically grounded and computationally efficient manner.

Building on Hierarchical Temporal Memory (HTM), the Cortical Learning Algorithm (CLA), and the Neural Association Algorithm (NAA), this work aims to design, implement, and evaluate an extended cortical architecture in which multiple unsupervised learning units form independent contextual representations and arrive at coherent interpretations through structured interaction rather than centralised control.

The comparative analysis presented in Section 3.4 and summarised in Table 1 informs this research aim by highlighting the absence of architectures that combine biological plausibility, continual learning, and higher-level contextual association within a single framework.

4.2 Research Questions

Guided by the literature reviewed in the preceding sections, the research is structured around the following core research questions:

1. How can contextual dependencies be encoded beyond first-order temporal transitions within cortical learning algorithms while preserving sparse, unsupervised, and biologically plausible learning principles?
2. Which architectural extensions enable the integration of hierarchical and heterarchical contextual information across multiple cortical learning units?
3. How can voting or consensus mechanisms among distributed cortical units support higher-level contextual association without introducing global error signals or supervised training?

4. How does the proposed heterarchical cortical architecture compare to existing deep neural network (DNN), spiking neural network (SNN), and CLA-based approaches in terms of learning efficiency, generalisation, continual learning, and biological plausibility?

4.3 Research Objectives

To address these research questions, the project is structured around the following objectives:

- To critically analyse existing cortical learning algorithms, including HTM-CLA and NAA, with respect to their ability to represent and utilise contextual information.
- To design a heterarchical cortical learning architecture composed of multiple unsupervised learning units capable of forming independent sparse distributed representations of context.
- To develop and implement voting or consensus mechanisms that allow distributed cortical units to combine their contextual inferences without reliance on supervised error signals.
- To evaluate the proposed architecture on representative sequence and context-modelling tasks, comparing its performance and behaviour against DNN-, SNN-, and CLA-based baselines.
- To assess the proposed model in terms of biological plausibility, computational efficiency, robustness to noise, and suitability for continual learning.
- To provide empirical evidence that heterarchical organisation and distributed voting contribute to improved contextual association beyond existing cortical learning models.

It is noted that the research objectives are addressed across different stages of the thesis. The first objective, which concerns the critical analysis of existing cortical learning algorithms and related approaches, is primarily addressed in Chapter 3 through the literature review and comparative analysis. The remaining objectives focus on the design, implementation, and evaluation of extended cortical learning architectures and therefore form the basis of the subsequent methodological, experimental, and analytical chapters.

5 Methodology

The methodological approach adopted in this research is directly informed by the findings of the literature review presented in Chapter 3 and the research gap identified in

Chapter 4. As shown in Sections 3.2 and 3.3, existing cortical learning algorithms and associative extensions provide biologically grounded mechanisms for sparse, unsupervised, and continual learning, but lack explicit architectural support for higher-level contextual integration across multiple learning units. Section 3.4 further demonstrates that alternative computational paradigms, while effective in capturing higher-order dependencies, rely on dense representations and global optimisation procedures that are incompatible with the biological and continual learning constraints motivating this work.

Given these observations, a computational and experimental methodology based on architectural extension and comparative evaluation is appropriate. The emphasis on modular architectural design reflects the limitations of single-region cortical models identified in Sections 3.2 and 3.3, while the use of controlled synthetic experiments aligns with prior work on analysing contextual ambiguity and representational behaviour in cortical-inspired systems. The choice to focus on continual, unsupervised learning under streaming input conditions follows directly from the defining properties of cortical learning algorithms and addresses the shortcomings of offline, task-specific optimisation highlighted in Section 3.4.

Accordingly, the methodology is structured to (i) build upon established cortical learning components as baseline reference models, (ii) explore heterarchical architectural extensions motivated by limitations identified in prior work, and (iii) evaluate behavioural and representational properties relevant to contextual association rather than benchmark performance alone. This alignment ensures that the proposed methodological steps are grounded in the existing literature and directly address the research gap motivating the study.

5.1 Overall Research Design

This research adopts a computational and experimental methodology grounded in biologically inspired modelling. The project is structured around the design, implementation, and evaluation of extended cortical learning architectures that build upon existing Cortical Learning Algorithms (CLA) and the Neural Association Algorithm (NAA). The methodology follows an iterative cycle of theoretical analysis, algorithmic development, empirical evaluation, and refinement, ensuring that architectural choices are informed both by neuroscience literature and by observed computational behaviour.

Rather than treating learning as a static optimisation problem, the methodology emphasises continual, unsupervised learning under streaming input conditions. This design choice reflects the operating constraints of biological cortex and directly aligns with the research aim of modelling higher-level contextual association without reliance on global error signals or labelled supervision.

5.2 Baseline Models and Reference Architectures

As a foundation, existing implementations of HTM-CLA and NAA are used as baseline models. HTM-CLA provides established mechanisms for sparse distributed representation, temporal sequence learning, and prediction via the Spatial Pooler and Temporal Memory components. NAA extends this framework by introducing associative mechanisms based on dendritic processing and structural plasticity.

These baseline models serve two methodological purposes. First, they provide a validated point of comparison against which architectural extensions can be evaluated. Second, they act as modular building blocks that can be composed, modified, or replicated to explore heterarchical organisation and inter-unit interaction.

5.3 Proposed Architectural Extensions

To address the absence of explicit multi-unit contextual integration identified in Chapter 3, the central methodological contribution of this research is the design of a heterarchical cortical learning architecture composed of multiple independent learning units. Each unit operates as an unsupervised cortical module capable of forming sparse distributed representations and learning context-sensitive associations from its own input stream.

As discussed in Sections 3.2 and 3.3, existing cortical learning algorithms primarily encode context through temporal adjacency within individual or weakly coupled learning units, and do not provide explicit mechanisms for integrating semantic, spatial, or functional context across multiple modules. Section 3.4 further shows that alternative computational approaches address higher-level dependencies at the cost of biological plausibility and continual learning. The extensions introduced here are therefore designed to address these specific limitations while preserving the defining properties of cortical learning algorithms.

Rather than enforcing a strict hierarchical flow of information, the architecture supports heterarchical connectivity, allowing learning units to exchange contextual signals laterally and through feedback pathways. A key design element is the introduction of voting or consensus mechanisms, through which multiple learning units contribute to higher-level contextual interpretation. These mechanisms are implemented without global optimisation, relying instead on local confidence measures, SDR overlap, or predictive agreement.

This design choice directly addresses the lack of multi-area coordination identified in Section 3.2.4 and the absence of heterarchical integration mechanisms discussed in Section 3.3.3. The design space explored includes variations in the number of learning units, connectivity patterns, and voting strategies. These variations allow systematic investigation of how distributed contextual association emerges under different architectural constraints.

5.4 Implementation Strategy

The models are implemented in a modular software framework that allows individual components (e.g., SDR encoders, associative memory, voting modules) to be developed and tested independently. Existing open-source implementations of HTM-CLA and NAA are extended where appropriate, while custom modules are developed to support heterarchical interaction and voting.

All learning is performed online, with identical mechanisms used during training and inference. Input data are presented as continuous streams rather than fixed datasets, ensuring that learning dynamics reflect continual adaptation rather than offline optimisation. Where relevant, controlled synthetic datasets are used to isolate specific contextual properties (e.g., temporal ambiguity, semantic overlap), while real-world datasets are employed to assess robustness and scalability.

5.5 Evaluation Methodology

Evaluation focuses on qualitative and quantitative measures relevant to contextual association. These include prediction accuracy under contextual ambiguity, stability of learned representations over time, robustness to noise and distributional shift, and the ability to generalise learned context to novel but related inputs.

Comparative evaluation is conducted against representative deep neural network and spiking neural network models, as well as baseline CLA/NAA configurations. Rather than optimising for benchmark performance alone, emphasis is placed on behavioural characteristics such as continual learning capability, interpretability of representations, and sensitivity to contextual cues. This evaluation strategy reflects the research objective of understanding *how* context is learned and used, not merely *how well* a model performs on a specific task.

6 Results and Discussion: Progress to Date

6.1 Scope and Nature of Current Results

This section defines the assumptions, constraints, and design considerations that inform the architectural and implementation work planned for the subsequent phases of the PhD project. These considerations are derived from the research framework and methodology established in Chapters 3 to 5. This provides the basis for the architectural design and evaluation activities associated with Work Packages 7 and 8, which were originally defined in RDC.1 and are reproduced here in Chapter 8 while discussing the current progress of the project. The factors outlined here delimit the scope within which heterarchical integration mechanisms will be specified, implemented, and assessed.

6.2 Analysis of Baseline Cortical Learning Models

To ground the proposed research in existing practice and to assess the practical realisation of cortical learning concepts described in the literature, a detailed examination of representative baseline implementations was conducted. This analysis focused on three publicly available codebases: the *nupic.core-legacy* [25] implementation of Hierarchical Temporal Memory and Temporal Memory models, the *neocortexapi* [7] implementation of the Neural Association Algorithm, and early reference implementations associated with the Thousand Brains Theory [34]. Together, these implementations represent the most mature and influential attempts to operationalise cortical learning principles in software.

The *nupic.core-legacy* repository was examined as a reference implementation of HTM and Temporal Memory models. The analysis focused on the architectural assumptions underlying sequence learning, the representation of sparse distributed representations, and the role of dendritic segments in predictive state transitions. The implementation robustly supports online, unsupervised learning and local synaptic adaptation driven by temporal input streams. Contextual modulation is realised primarily through distal dendritic activity and cell state transitions, enabling prediction of immediate future inputs based on recent history. However, the architectural scope of the implementation is largely confined to a single cortical region abstraction. While the codebase supports multiple regions at a structural level, there is no explicit mechanism for structured interaction or coordination between regions to support higher-level contextual association. Context remains predominantly temporal and local, emerging from sequence transitions rather than from interaction across distributed learning units.

The *neocortexapi* repository was examined as a concrete implementation of the Neural Association Algorithm. Particular attention was paid to the representation of canonical units, dendritic segments, and the mechanisms used to associate multiple input modalities or contextual signals within a unit. The implementation extends core HTM concepts by supporting associative binding and structural plasticity at the level of individual canonical units, allowing richer context representation than pure sequence memory. The examined architecture enables local integration of multiple contextual signals and supports adaptive association through unsupervised learning. Nevertheless, the design largely assumes that associative processing occurs within isolated or weakly coupled units. Interactions between multiple canonical units are limited and do not provide an explicit architectural mechanism for coordinated inference or context integration across a population of units. As a result, higher-level contextual association remains implicit and localised rather than emerging from structured inter-unit interaction.

Early implementations associated with the Thousand Brains Theory were examined to assess how multi-model inference and distributed context representation are addressed in practice. The analysed code reflects an exploratory architectural direction in which multiple cortical models operate in parallel, often grounded in sensorimotor reference frames. These implementations explore how individual models can learn object-centric

representations and contribute partial hypotheses to a collective inference process. While this represents a significant conceptual shift towards distributed representation and multi-model interaction, the examined codebases remain experimental in nature. Architectural abstractions are still evolving, coordination mechanisms between models are task-specific, and there is no unified framework for integrating semantic, spatial, and functional context across models in a generalised manner. Consequently, while the Thousand Brains implementations point towards heterarchical organisation, they stop short of providing a formalised and reusable architectural mechanism for higher-level contextual association.

Across all three implementations, several common characteristics were observed. All rely on sparse distributed representations, local synaptic plasticity, and continual online learning without global optimisation or supervised error signals. These shared properties align closely with biological constraints and reinforce their suitability as baselines for biologically inspired learning research. At the same time, a consistent limitation across implementations is the absence of explicit architectural support for integrating contextual information across multiple learning units or regions. Contextual association is either confined to temporal adjacency, realised locally within a single unit, or handled implicitly through task-specific coordination logic.

The outcome of this analysis is twofold. First, it confirms that existing baseline implementations provide a strong foundation for local sequence learning and associative processing under biological constraints. Second, it highlights a clear architectural gap: none of the examined systems offer a principled mechanism for higher-level contextual association emerging from structured interaction between multiple learning units operating in parallel. This observation directly informs the assumptions and constraints articulated in this chapter and motivates the proposed focus on heterarchical integration as a core architectural extension in the subsequent phases of the research.

6.3 Discussion and Implications for Further Work

The analysis presented in this report yields several findings that shape the direction of the proposed PhD research. While no architectural implementation or empirical evaluation has yet been undertaken, the combined literature review and examination of existing implementations provide clear insights into the current state of cortical learning models and their limitations with respect to higher-level contextual association.

First, the literature review and implementation analysis confirm that existing cortical learning algorithms are effective at modelling local sequence learning and temporal context under biological constraints. Both HTM-based Temporal Memory models and associative extensions rely on sparse distributed representations, local synaptic plasticity, and continual online learning. However, across both theoretical models and reference implementations, contextual information is predominantly encoded through temporal adjacency or local dendritic modulation within individual learning units. There is no explicit architectural mechanism for integrating contextual information across multiple units or

regions in a coordinated manner.

Second, associative extensions such as the Neural Association Algorithm demonstrate that richer contextual representations can be learned within a single canonical unit. The examination of existing implementations shows that such mechanisms enable multi-modal and associative binding at a local level. Nevertheless, these approaches do not address the problem of distributed contextual inference, as interactions between multiple units remain implicit or weakly defined. As a result, higher-level context does not emerge from structured inter-unit interaction.

Third, exploratory implementations associated with the Thousand Brains Theory highlight the potential importance of distributed and heterarchical organisation in cortical learning. These implementations suggest that parallel models operating over different reference frames may contribute complementary hypotheses. However, the absence of a formalised architectural framework limits their applicability as a general solution for higher-level contextual association.

Taken together, these findings indicate that the primary challenge is not the absence of biologically plausible learning mechanisms, but the lack of architectural structures that enable coordinated contextual integration across multiple learning units. This observation motivates the focus of the proposed research on heterarchical integration as a means of supporting higher-level contextual association while preserving the defining properties of cortical learning algorithms.

The implications for further work are therefore centred on architectural design rather than algorithmic optimisation. Subsequent phases of the research will focus on specifying and implementing mechanisms for structured interaction between multiple learning units, defining how contextual information is combined across units, and evaluating whether such architectures support semantic, spatial, and functional context beyond temporal association alone. These activities form the basis for the planned implementation and evaluation work following confirmation of route.

7 Statement of Originality and Contribution to Knowledge

The contribution of this PhD research is positioned at the level of architectural principles and representational mechanisms for higher-level contextual association in biologically inspired cortical learning systems. At the RDC.2 stage, the contributions presented in this report are primarily analytical and conceptual, establishing the basis for subsequent architectural design, implementation, and evaluation.

7.1 Originality of the Research Focus

The literature review demonstrates that cortical learning algorithms provide biologically grounded mechanisms for sparse distributed representations, local learning, and continual online adaptation (Sections 3.2 and 3.3). However, the review also shows that existing

models primarily encode context through temporal adjacency and local dendritic modulation, with limited support for integrating semantic, spatial, and functional context across multiple learning units (Sections 3.2.3, 3.3.3). In parallel, alternative computational paradigms such as deep neural networks and spiking neural networks offer complementary strengths but do not satisfy the combined constraints of biological plausibility and continual learning that motivate this work (Section 3.4).

The originality of the research focus lies in addressing this gap directly: the PhD targets the design of cortical-inspired architectures in which higher-level contextual association emerges from structured interaction between multiple learning units, rather than being confined to temporal sequence memory or local association within a single unit. This focus is defined explicitly as a research gap in Chapter 4 and is presented as a specific architectural deficiency identified across the reviewed paradigms (Section 4.1).

7.2 Contribution to Knowledge at RDC.2

Although implementation and empirical evaluation are planned for subsequent phases, the work completed to date yields contributions to knowledge in three concrete forms.

(1) A synthesised and traceable definition of the research gap for higher-level contextual association in cortical learning models. This report consolidates findings from cortical learning algorithms (Section 3.2), associative extensions such as NAA (Section 3.3), and comparative paradigms including deep and spiking neural networks (Section 3.4) into a single, explicitly stated research gap (Section 4.1). The gap is defined in architectural terms: existing biologically inspired models lack explicit mechanisms for distributed contextual integration across multiple learning units required for higher-level contextual association. This synthesis provides a structured basis for research direction that goes beyond isolated limitations described in individual strands of prior work.

(2) An analysis of baseline implementations and their architectural assumptions. In addition to reviewing conceptual models, this report examines representative baseline implementations used in practice, including reference HTM/Temporal Memory implementations and publicly available implementations of NAA and early work on distributed and modular sensorimotor learning (Section 6.2). This analysis contributes architectural insight that is not available from the literature alone by clarifying how context is operationalised in existing systems (predominantly local and temporal) and by identifying the practical absence of generalised mechanisms for coordinated inference across multiple learning units. These findings directly constrain and inform subsequent architectural design.

(3) A methodology aligned to the identified gap and constrained by biological plausibility and continual learning. Building on the reviewed literature and imple-

mentation analysis, this report defines a methodological approach explicitly aligned with the research gap (Section 5.1) and introduces a set of architectural extensions motivated by documented limitations of prior work (Section 5.3). The methodology is formulated to preserve key biological constraints i.e. sparse representations, local learning, and online continual adaptation, while enabling investigation of distributed contextual integration. The contribution at this stage is the definition of a justified and traceable research programme suitable for developing and evaluating heterarchical integration mechanisms in subsequent phases.

7.3 Planned Contribution Beyond RDC.2

The contributions presented in this report establish the foundation for the planned research contributions of the PhD. Following confirmation of route, the work will focus on the formal architectural specification of heterarchical integration mechanisms, prototype implementation, and empirical evaluation using controlled experimental scenarios designed to test higher-level contextual association (Chapters 5 and 8). The intended outcome of this subsequent work is to demonstrate whether distributed cortical-inspired architectures can represent and integrate semantic, spatial, and functional context beyond temporal association while maintaining biological plausibility and continual learning constraints.

8 Plan of Research and Timeline

This chapter presents the research plan and timeline in relation to the work packages defined in the RDC.1 submission and provides an updated view reflecting progress achieved to date. The plan is structured around specific work packages that define the scope, focus, and sequencing of the PhD research, allowing direct mapping between completed work, current status at RDC.2, and planned activities.

8.1 Defined Work Packages

The PhD research is structured around a set of work packages that define the scope, sequencing, and progression of the project. These work packages were originally defined at the RDC.1 stage and are reproduced here to provide a complete and self-contained overview of the research plan and current progress.

Work Packages	Focus	Status	Reflected in RDC.2
WP1–WP2	Literature review of cortical learning algorithms, neural association models, and contextual association mechanisms	Completed	Chapters 2 and 3
WP3	Definition of research direction and planning (RDC.1)	Completed	Chapter 4 (refined aims, gap, objectives)
WP4, WP6	Comparative and critical analysis of CLAs, DNNs, and SNNs; identification of limitations	Completed	Sections 3.3 and 3.4
WP7	Requirements definition and scope identification for higher-level contextual association; identification of heterarchical direction	Substantially completed	Chapters 3, 4, and 5
WP8	Formal architectural design of contextual association mechanisms	Planned (09.25-02.26)	Chapter 5 (conceptual basis)
WP10–WP12	Prototype implementation, testing, and evaluation of contextual association mechanisms	Planned (02.26-06.27)	–
WP13	Generalisation of contextual association mechanisms and dissemination	Planned (07.27-09.27)	–
WP14	Writing and submission of PhD thesis (RDC.3)	Planned (08.27-02.28)	–

Table 2: Summary of Work Packages and Research Progress

8.2 Progress Against the Original Plan

The research conducted up to the RDC.2 milestone corresponds primarily to Work Packages 1–6 defined in the RDC.1 work plan. These work packages focused on establishing the conceptual and analytical foundation for the research.

Work Packages 1 and 2 addressed the literature review of cortical learning algorithms, neural association models, and contextual association mechanisms. This work is reflected in Chapters 2 and 3 of this report, which review cortical learning algorithms, the Neural Association Algorithm, neuroscientific perspectives on context, and alternative computational approaches including deep and spiking neural networks.

Work Package 3 involved the formulation of the initial research direction and planning through the RDC.1 submission. The refinement and consolidation of this direction,

including clarification of the research gap, aims, and objectives, is presented in Chapter 4 of this report.

Work Packages 4 and 6 focused on comparative analysis and critical assessment of existing approaches to contextual association. These activities are reflected in Sections 3.3 and 3.4, including the comparative evaluation of cortical learning algorithms, deep neural networks, and spiking neural networks, and the identification of their respective limitations with respect to higher-level contextual integration.

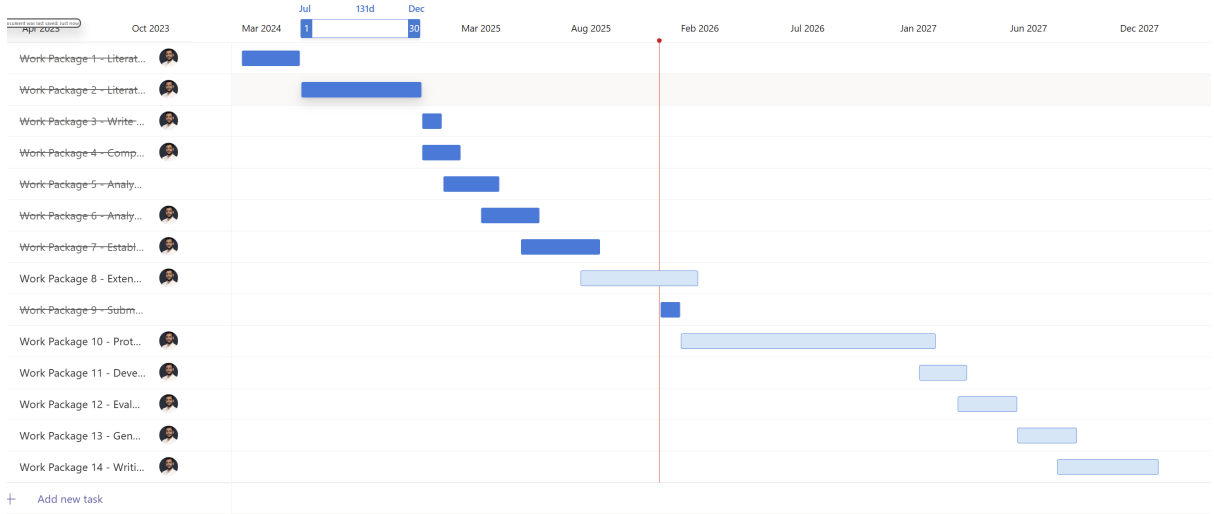


Figure 1: Workplan progress

Collectively, these completed work packages establish the analytical and conceptual basis required to proceed with architectural design and implementation work following confirmation of route.

8.3 Updated Research Plan

Following confirmation of route, the remaining research will proceed through a revised sequence of work packages derived from the RDC.1 plan and refined based on insights gained during the literature review and comparative analysis.

The next phase of work will focus on requirements definition and architectural design for higher-level contextual association in cortical learning algorithms (corresponding to Work Packages 7 and 8). This includes defining architectural mechanisms for integrating semantic, spatial, and functional context across multiple learning units and specifying representational and interaction principles consistent with biological plausibility and continual learning.

Subsequent phases will address prototype implementation and validation (Work Packages 10–12). This work will involve the development of a proof-of-concept implementation, the design of targeted test scenarios and use cases to evaluate contextual association

behaviour, and systematic evaluation of the proposed architecture under controlled experimental conditions. Based on the findings of the literature review and comparative analysis, the scope of the research was refined to include heterarchical integration between multiple learning units as a central focus for subsequent implementation and evaluation.

The final stages of the research will focus on generalisation and dissemination (Work Package 13) and the preparation and submission of the PhD thesis (Work Package 14). The overall timeline remains aligned with the original RDC.1 plan, with adjustments reflecting the clarified research scope established at RDC.2.

8.4 Feasibility and Timeline

The timeline defined in the original workplan remains realistic and achievable within the remaining registration period. Later work packages focus on implementation, evaluation, and thesis writing, with sufficient overlap to allow iterative refinement of both experimental work and written material. The staged nature of the plan ensures that risks associated with implementation complexity are mitigated through early prototyping and incremental evaluation.

9 Data Management Plan

This research primarily involves computational modelling, algorithm development, and experimental evaluation using synthetic, simulated, and selected publicly available datasets. No personal, sensitive, or confidential data are collected or processed as part of the project. The Data Management Plan therefore focuses on the handling of code, experimental configurations, generated results, and derived datasets produced during the research.

9.1 Types of Data

The data generated and used in this project include:

- Source code implementing cortical learning algorithms, architectural extensions, and experimental frameworks.
- Synthetic and mock datasets designed to test specific contextual properties (e.g. ambiguity, temporal dependencies, partial observability).
- Derived data such as model outputs, prediction logs, performance metrics, and visualisations produced during experimental evaluation.
- Documentation describing experimental setups, parameter configurations, and evaluation protocols.

Where external datasets are used, these will be limited to publicly available sources suitable for non-commercial academic research and will be used in accordance with their respective licences.

9.2 Data Storage and Backup

All research data and source code are stored digitally on secure, password-protected systems. Primary storage is maintained on institutional or personal development environments, with regular backups to encrypted cloud-based storage and version-controlled repositories. Source code and configuration files are managed using distributed version control systems to ensure traceability, reproducibility, and protection against data loss.

Intermediate experimental outputs are stored in structured directory formats with clear naming conventions and metadata descriptions. Redundant or obsolete data generated during exploratory experimentation are periodically reviewed and removed to minimise unnecessary storage.

9.3 Data Access, Sharing, and Preservation

In line with open research practices, non-proprietary source code, experimental scripts, and selected derived datasets will be made available in public repositories (e.g. Git-based platforms) where feasible and appropriate. Access will be provided subject to institutional guidelines and intellectual property considerations.

Data required to support the main findings of the thesis will be preserved for a minimum period in accordance with University of Plymouth policies. Documentation will be provided to enable reuse and replication of experiments by other researchers. Where full data sharing is not practical, sufficient methodological detail will be included in the thesis to ensure transparency and reproducibility.

9.4 Ethics, Intellectual Property, and Legal Considerations

This research does not involve human participants, personal data, or animal experimentation and therefore does not require formal ethical approval. All data generated are computational in nature.

Intellectual property arising from the research will be governed by University of Plymouth regulations. Any third-party software or datasets used will be properly acknowledged, and their licence conditions respected. No commercial or restricted data sources are anticipated.

9.5 Responsibilities

The researcher is responsible for the day-to-day management of research data, including storage, backup, documentation, and compliance with institutional policies. Oversight

and guidance are provided by the supervisory team to ensure that data management practices remain appropriate throughout the duration of the project.

10 Conclusion

This RDC.2 report is situated within the broader area of biologically inspired learning systems and computational models of the neocortex. The focus of the PhD research is on cortical learning algorithms and their ability to support contextual association, with particular emphasis on the representation and integration of temporal, semantic, spatial, and functional context.

Through a critical review of the literature presented in Chapter 3, this report examined cortical learning algorithms and associative extensions (Sections 3.2 and 3.3), as well as alternative computational approaches including deep neural networks and spiking neural networks (Section 3.4). This analysis identified limitations related to single-region learning, temporal-only context encoding, and the absence of explicit mechanisms for distributed contextual integration. These limitations were synthesised into a clearly defined research gap in Section 4.1, which motivates the focus on heterarchical integration across multiple learning units.

The proposed research adopts a computational approach grounded in biologically inspired modelling through the use of sparse distributed representations, local synaptic plasticity mechanisms, and dendritic processing to incorporate contextual modulation. Learning is performed in an online and continual manner from streaming input, without reliance on global error backpropagation or task-specific supervision, reflecting key organisational and functional principles observed in cortical systems. Initial exploratory work, including small-scale prototyping of heterarchical voting mechanisms using synthetic data, has been undertaken to assess feasibility and inform architectural design decisions. While these early investigations do not constitute empirical validation, they provide a concrete foundation for the systematic experimental evaluation planned in the next stages of the research.

The aims, research questions, and objectives defined in this report establish a clear and achievable scope for the remainder of the PhD. The proposed programme of work, building on the timeline established at RDC.1 and refined here, provides a structured pathway toward thesis completion. Data management and ethical considerations have been addressed in a manner appropriate to the computational nature of the project.

Overall, this RDC.2 report aims to demonstrate that the research direction is well defined, methodologically grounded, and feasible within the remaining registration period. The work completed to date has clarified the research problem and established a coherent strategy for addressing it, supporting progression to the next phase of the PhD and the development of the proposed contributions to knowledge.

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